

with (LinearAlgebra);

[&x, Add, Adjoint, BackwardSubstitute, BandMatrix, Basis, BezoutMatrix, BidiagonalForm, (1)
 BilinearForm, CARE, CharacteristicMatrix, CharacteristicPolynomial, Column, ColumnDimension,
 ColumnOperation, ColumnSpace, CompanionMatrix, CompressedSparseForm, ConditionNumber,
 ConstantMatrix, ConstantVector, Copy, CreatePermutation, CrossProduct, DARE, DeleteColumn,
 DeleteRow, Determinant, Diagonal, DiagonalMatrix, Dimension, Dimensions, DotProduct,
 EigenConditionNumbers, Eigenvalues, Eigenvectors, Equal, ForwardSubstitute, FrobeniusForm,
 FromCompressedSparseForm, FromSplitForm, GaussianElimination, GenerateEquations,
 GenerateMatrix, Generic, GetResultDataType, GetResultShape, GivensRotationMatrix,
 GramSchmidt, HankelMatrix, HermiteForm, HermitianTranspose, HessenbergForm, HilbertMatrix,
 HouseholderMatrix, IdentityMatrix, IntersectionBasis, IsDefinite, IsOrthogonal, IsSimilar, IsUnitary,
 JordanBlockMatrix, JordanForm, KroneckerProduct, LA_Main, LUDecomposition, LeastSquares,
 LinearSolve, LyapunovSolve, Map, Map2, MatrixAdd, MatrixExponential, MatrixFunction,
 MatrixInverse, MatrixMatrixMultiply, MatrixNorm, MatrixPower, MatrixScalarMultiply,
 MatrixVectorMultiply, MinimalPolynomial, Minor, Modular, Multiply, NoUserValue, Norm,
 Normalize, NullSpace, OuterProductMatrix, Permanent, Pivot, PopovForm, ProjectionMatrix,
 QRDecomposition, RandomMatrix, RandomVector, Rank, RationalCanonicalForm,
 ReducedRowEchelonForm, Row, RowDimension, RowOperation, RowSpace, ScalarMatrix,
 ScalarMultiply, ScalarVector, SchurForm, SingularValues, SmithForm, SplitForm,
 StronglyConnectedBlocks, SubMatrix, SubVector, SumBasis, SylvesterMatrix, SylvesterSolve,
 ToeplitzMatrix, Trace, Transpose, TridiagonalForm, UnitVector, VandermondeMatrix, VectorAdd,
 VectorAngle, VectorMatrixMultiply, VectorNorm, VectorScalarMultiply, ZeroMatrix, ZeroVector, Zip]

with (LinearAlgebra[Generic]);

[BareissAlgorithm, BerkowitzAlgorithm, CharacteristicPolynomial, Determinant, GaussianElimination, (2)
 GenericCheck, HermiteForm, HessenbergAlgorithm, HessenbergForm, LinearSolve, MatrixAdd,
 MatrixInverse, MatrixMatrixMultiply, MatrixVectorMultiply, MinorExpansion, NullSpace, RREF,
 ReducedRowEchelonForm, SmithForm, StronglyConnectedBlocks, VectorAdd]

q := 11;

q := 11 (3)

n := 3;

n := 3 (4)

Pmod := x^n + 1;

Pmod := x^3 + 1 (5)

Zqm['='] := **proc**(a, b) evalb(rem(a, Pmod, x) **mod** q = rem(b, Pmod, x) **mod** q); **end proc**;

Zqm._ := **proc**(a, b) evalb(rem(a, Pmod, x) **mod** q = rem(b, Pmod, x) **mod** q) **end proc** (6)

Zqm['0'] := 0;

Zqm_0 := 0 (7)

Zqm['1'] := 1;

Zqm_1 := 1 (8)

Zqm['*'] := **proc**() **local** y; rem(mul(y, y = _passed[]), Pmod, x) **mod** q; **end proc**;

Zqm._* := **proc**() **local** y; rem(mul(y, y = args[]), Pmod, x) **mod** q **end proc** (9)

Zqm['+'] := **proc**() **local** y; rem(add(y, y = _passed[]), Pmod, x) **mod** q; **end proc**;

Zqm._+ := **proc**() **local** y; rem(add(y, y = args[]), Pmod, x) **mod** q **end proc** (10)

Zqm['-'] := **proc**(a, b) rem(a - b, Pmod, x) **mod** q;

end proc;

$$Zqm_ := \text{proc}(a, b) \text{ rem}(a - b, Pmod, x) \text{ mod } q \text{ end proc} \quad (11)$$

```

Zqm[\' \'] := proc(a, b)
  local i, d; d := Gcdex(b, Pmod, x, 'i') mod q;
  if degree(d, x) ≠ 1 then error "inverse not defined";
  end if;
  rem(i * a, Pmod, x) mod q;
end proc;

```

$$Zqm_ := \text{proc}(a, b) \quad (12)$$

```

  local i, d;
  d := Gcdex(b, Pmod, x, 'i') mod q;
  if degree(d, x) <> 1 then error "inverse not defined" end if;
  rem(i * a, Pmod, x) mod q

```

end proc

$Ax := \text{Matrix}([[5 + 2x, 9x^2 + x + 5]]);$

$$Ax := \begin{bmatrix} 5 + 2x & 9x^2 + x + 5 \end{bmatrix} \quad (13)$$

$Bx := \text{Matrix}([[10x^2 + x + 9]]);$

$$Bx := \begin{bmatrix} 10x^2 + x + 9 \end{bmatrix} \quad (14)$$

$s := \text{Matrix}([[x + 1], [x - 1]]);$

$$s := \begin{bmatrix} x + 1 \\ x - 1 \end{bmatrix} \quad (15)$$

$buff := \text{MatrixMatrixMultiply}[Zqm](Ax, s);$

$$buff := \begin{bmatrix} 5x^2 + 2 \end{bmatrix} \quad (16)$$

$buff := \text{MatrixAdd}[Zqm](Bx, -buff);$

$$buff := \begin{bmatrix} 5x^2 + x + 7 \end{bmatrix} \quad (17)$$

Since 1 is closer to q we can deduct that $m_1 = 0$, On the other hand, 5 and 7 are closer to $\frac{q}{2}$, therefore m_2 and $m_0 = 1$. Hence the message is $X^2 + 1$.

Name: Diya Jain U81673988